Because different observers perceive that the interaction takes place at various locations along the left part of the surface of Figure 6.10, in a real sense this means that the interaction location is smeared out among all of them. This spreads out the force's punch and, in the case of the gravitational force, this smearing significantly dilutes its ultramicroscopic properties—so much so that calculations yield well-behaved finite answers in place of the previous infinities. This is a more precise version of the smearing encountered in the rough answer of the last section. And once again, this smearing results in a smoothing of the ultramicroscopic jitteriness of space as sub-Planck-length distances are blurred together.

Like viewing the world through glasses that are too weak or too strong, fine sub-Planckian details that would be accessible to a point-particle probe are smeared together by string theory and rendered harmless. And unlike the case with poor eyesight, if string theory is the ultimate description of the universe, there is no corrective lens to bring the supposed sub-Planck-scale fluctuations into sharp focus. The incompatibility of general relativity and quantum mechanics—which would become apparent only on sub-Planck-scale distances—is avoided in a universe that has a lower limit on the distances that can be accessed, or even said to exist, in the conventional sense. Such is the universe described by string theory, in which we see that the laws of the large and the small can be harmoniously merged together as the supposed catastrophe arising on ultramicroscopic distances is summarily done away with.

Beyond Strings?

Strings are special for two reasons. First, even though they are spatially extended they can be described consistently in the framework of quantum mechanics. Second, among the resonant vibrational patterns there is one that has the exact properties of the graviton, thus ensuring that the gravitational force is an intrinsic part of its structure. But just as string theory shows that the conventional notion of zero-dimensional point particles appears to be a mathematical idealization that is not realized in the real world, might it also be the case that an infinitely thin one-dimensional strand is similarly a mathematical idealization? Might it actually be the case that strings have some thickness—like the surface of a two-dimensional bicycle-tire inner